

PROJECT REPORT

FinWise AI

Intelligent Loan Eligibility, Credit Analysis & Financial Advisory Platform

SmartBridge Vibe Coding Internship (SIP 2026) - SRM University AP

Team Members	Role
Jainesh Bharti	Frontend Development + AI Integration
Navneeth Biyyapu	Frontend Development + Google Sheets Integration
Vishnu Vardhana Reddy Paluvai	Frontend Development + UI Development

1. Introduction

1.1 Project Overview

FinWise AI is a smart, AI-powered BFSI (Banking, Financial Services & Insurance) web platform built to simplify financial decision-making. It provides users with an intelligent loan-eligibility checker, a credit-score analyzer, an EMI calculator, and AI-generated financial guidance - all within a single, responsive web interface, with no login required. It is deployed as a static site on GitHub Pages and uses Groq AI and Google Sheets (via Apps Script) as its only external services.

1.2 Purpose

The purpose of FinWise AI is to give individuals - especially first-time or under-banked borrowers - a free, instant, and easy-to-understand way to assess their loan eligibility, understand their credit standing, calculate accurate EMIs, and receive personalised financial advice, before they approach a bank or NBFC.

2. Ideation Phase

2.1 Problem Statement

PS No.	I am	Trying to	But	Because	Which makes me feel
PS-1	a salaried individual planning to apply for a loan	find out whether I am eligible before applying	banks take days to respond, or reject me after a hard credit check	I don't know my eligibility criteria in advance	anxious and discouraged
PS-2	a first-time borrower with thin credit history	understand my credit score and its impact	there is no free tool that classifies and explains my credit profile	credit bureaus/banks don't provide plain-language advisory	confused, lacking confidence

2.2 Empathy Map Canvas

Dimension	Details
Says / Thinks	"Am I even eligible?" - worried a rejection will hurt their credit score further; unsure which numbers matter most.

Dimension	Details
Does / Feels	Searches loan eligibility online, visits multiple bank sites, delays applying; feels anxious until given a clear result.
Pains / Gains	Pain: no single free tool for eligibility + credit + EMI + advice. Gain: instant, clear, personalised results building confidence.

2.3 Brainstorming

The team grouped ideas into four categories - Loan & Credit Tools, AI-Powered Advisory, Data & Automation, and UI/UX - then prioritised by impact vs. effort. Four ideas were rated highest priority (P0): the Loan Eligibility Checker, Credit Score Analyzer, EMI Calculator, and Groq AI-powered Financial Tips, with Google Sheets logging as a supporting (P1) feature.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	User Action	Experience
Awareness	User needs a loan / wants to check finances	Searches online for eligibility/credit tools
Consideration	Lands on FinWise AI landing page	Sees clear feature cards: Eligibility, Credit Score, EMI, AI Tips
Interaction	Fills in the Eligibility / Credit Score / EMI form	Gets an instant, rule-based result with no signup
AI Guidance	Opens AI Financial Tips	Receives a personalised recommendation from Groq AI within seconds
Follow-up	Result is logged to Google Sheets in the background	User can revisit anytime; team can analyse aggregate trends

3.2 Solution Requirement (Functional & Non-Functional)

FR No.	Requirement (Epic)	Details
FR-1	User Data Entry	Enter salary, age, credit score, existing EMI with client-side validation
FR-2	Loan Eligibility Check	Rule-based check; eligible amount = Salary x 20
FR-3	Credit Score Analysis	Classify into Excellent / Good / Poor with tips
FR-4	EMI Calculator	Reducing-balance EMI formula with amortisation summary
FR-5	AI Financial Tips	Groq AI-generated personalised guidance
FR-6	Data Logging	POST results to Google Apps Script -> Google Sheets
NFR	Description	
Usability	Single-page, no-signup forms; instant results without reload	
Security	HTTPS delivery; no server-side storage of sensitive data	
Performance	Rule-based results in <1s; AI tips in a few seconds	
Availability	24x7 via GitHub Pages static hosting	

FR No.	Requirement (Epic)	Details
Scalability	Stateless frontend extendable with auth/backend later	

3.3 Data Flow Diagram

User -> FinWise AI Frontend (validates input, runs rule engine) -> [Groq AI API for tips] and [Google Apps Script -> Google Sheets for logging] -> Result rendered back to the user instantly.

3.4 Technology Stack

Component	Technology
User Interface	HTML5, CSS3 (glassmorphism), Vanilla JavaScript
Application Logic	JavaScript rule engines: eligibility.js, credit-score.js, emi.js
AI Recommendation Engine	Groq AI Chat Completions API (LLaMA-based, free tier)
Data Storage	Google Sheets
Serverless Backend	Google Apps Script (Web App / doPost)
Hosting	GitHub Pages

4. Project Design

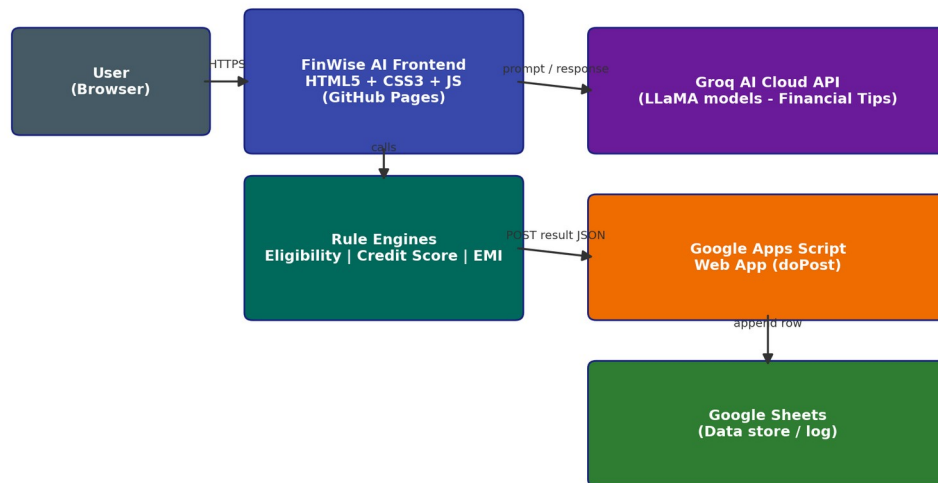
4.1 Problem - Solution Fit

Target users - first-time borrowers and salaried individuals - feel anxious and uninformed before applying for a loan, relying on scattered bank websites and generic advice. FinWise AI fits this behaviour by offering one instant, no-login destination that combines eligibility, credit and EMI checks with AI-generated guidance, leaving users confident and prepared.

4.2 Proposed Solution

Parameter	Description
Problem Statement	No free, instant way to check loan eligibility, credit standing or EMI before approaching a bank
Idea / Solution	A single-page platform combining Loan Eligibility, Credit Score Analysis, EMI Calculator and Groq AI Financial Tips
Novelty	Combines deterministic rule engines with generative-AI advice in one lightweight, no-login static app
Social Impact	Reduces rejection anxiety and helps under-banked users self-assess for free
Business Model	Freemium; potential lender-referral commissions or a premium credit-improvement report
Scalability	Can add auth, a real database, and more financial products over time

4.3 Solution Architecture



5. Project Planning & Scheduling

5.1 Project Planning

Sprint	Epic	Story	Story Points	Priority
Sprint-1	Loan Eligibility	USN-1	3	High
Sprint-1	Credit Score	USN-2	3	High
Sprint-1	EMI Calculator	USN-3	2	High
Sprint-2	AI Financial Tips	USN-4	5	Medium
Sprint-2	Data Logging	USN-5	3	Medium
Sprint-2	Responsive UI	USN-6	2	Low

Sprint-1: 8 story points completed (01-04 Jul 2026). Sprint-2: 10 story points completed (07-10 Jul 2026). Velocity = $18 / 2 = 9$ story points per sprint.

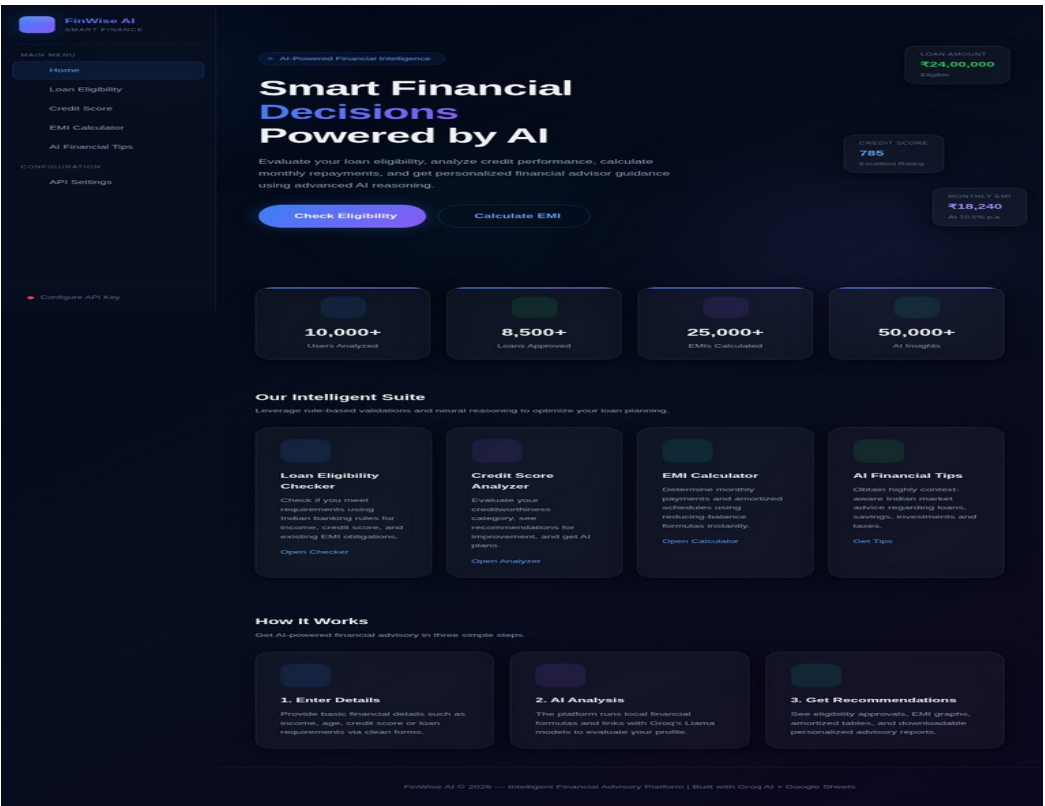
6. Functional and Performance Testing

6.1 Performance Testing

Test Case	Scenario	Result
FT-02	Loan eligibility rule evaluation	Pass - correct Eligible/Not Eligible + amount
FT-03	Credit score classification	Pass - correct label for all ranges
FT-04	EMI calculation accuracy	Pass - matched manual reducing-balance formula
FT-05	AI Tips generation (Groq)	Pass - relevant tips generated
FT-07	Google Sheets data logging	Pass - new row appended per submission
PT-01	Rule-based response time	Pass - under 1 second
PT-02	AI response time	Pass - ~2-4 seconds average

7. Results

7.1 Output Screenshots



Landing dashboard with quick eligibility, credit score and EMI snapshots.

FinWise AI

SMART FINANCE

MAIN MENU

Home

Loan Eligibility

Credit Score

EMI Calculator

AI Financial Tips

CONFIGURATION

API Settings

Configure API Key

Loan Eligibility Checker

Check your loan eligibility criteria and calculate eligible amounts instantly using rule-based thresholds.

Applicant Details

FULL NAME

John Doe

MONTHLY INCOME (₹)

₹ 45000

Must exceed ₹30,000 for eligibility

CREDIT SCORE (CIBIL)

750

Preferred score is greater than 700

EXISTING EMI OBLIGATIONS (₹)

₹ 5000

Must remain below ₹20,000

APPLICANT AGE

25

Must be at least 21 years old

Check Eligibility

ELIGIBILITY CRITERIA CHECKS

Monthly Income exceeds ₹30,000

Credit Score exceeds 700

Existing EMI below ₹20,000

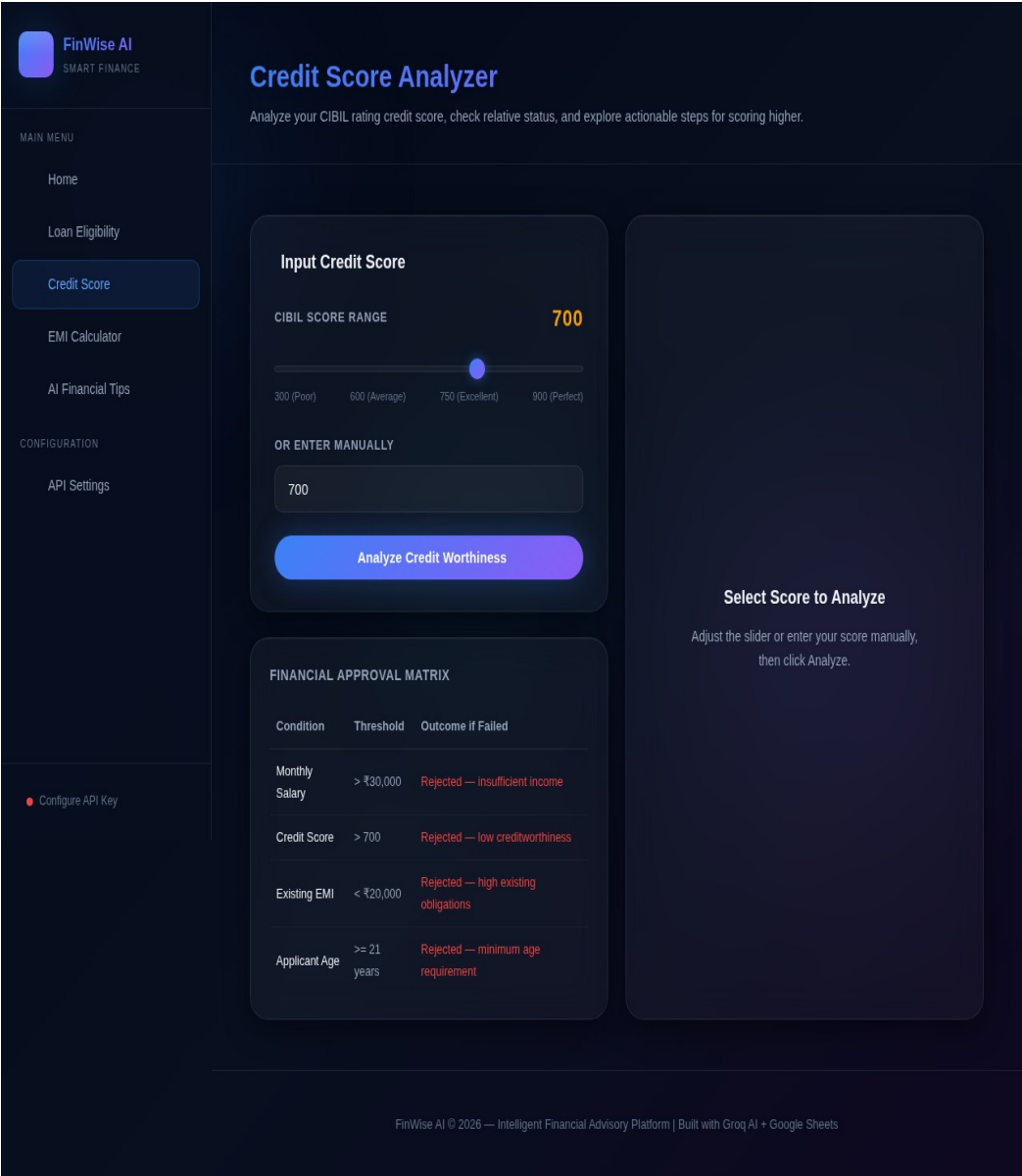
Applicant Age is 21 or older

Ready to Analyze

Fill out the form on the left with your financial details to check your eligibility status.

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Loan Eligibility Checker with real-time criteria checks.



Credit Score Analyzer.

FinWise AI

SMART FINANCE

MAIN MENU

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Credit Score

EMI Calculator

AI Financial Tips

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EMI Calculator

Compute monthly payments, track amortized splits, and evaluate loan metrics instantly.

Loan Parameters

PRINCIPAL LOAN AMOUNT (₹)

₹ 500000

ANNUAL INTEREST RATE (%)

10.5%

LOAN TENURE (MONTHS)

60

1 Yr (12M)

3 Yr (36M)

5 Yr (60M)

10 Yr (120M)

Calculate EMI

Standard Reducing Balance Formula

$$EMI = P \times R \times (1+R)^N / ((1+R)^N - 1)$$

Where: P = Principal, R = Monthly Rate (Annual Rate / 12 / 100), N = Tenure in Months.

Enter Details

Fill out the parameters on the left and click Calculate to view your monthly payments.

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EMI Calculator using the reducing-balance method.



FinWise AI

SMART FINANCE

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CONFIGURATION

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Configure API Key

AI Financial Tips

Consult our neural model for personalized investment strategies, debt reductions, and tax optimizations.

Your Financial Profile

MONTHLY INCOME (₹)

60000

CREDIT SCORE

750

EXISTING EMIS (₹)

5000

MONTHLY SAVINGS (₹)

15000

APPLICANT AGE

28

PRIMARY FINANCIAL GOAL

Purchase Home Loan

SPECIFIC QUESTION OR REQUEST

What are the best investment channels or loan prepayment options for my profile?

Generate AI Advisor Report

SUGGESTED ADVICE PROMPTS

Improve Loan Eligibility

Boost Credit Score

Reduce EMI Burden

Yield Investments

Consultation Report

Submit your profile context to generate customized advisory plans.

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AI Financial Tips powered by the Groq API.

8. Advantages & Disadvantages

Advantages:

- Free, instant, no-signup access to eligibility, credit and EMI checks
- Combines deterministic rule engines with generative-AI guidance in one app
- Lightweight static architecture - fast, cheap to host, and easy to maintain
- Mobile-responsive glassmorphism UI works across devices

Disadvantages:

- No server-side validation, so it complements rather than replaces a bank's formal assessment
- Requires the user to supply their own Groq API key for AI Tips
- No persistent user accounts, so history isn't tracked per individual user

9. Conclusion

FinWise AI successfully demonstrates how a lightweight, serverless web application can combine rule-based financial logic with generative AI to deliver a genuinely useful, free BFSI advisory tool. Built entirely with HTML5, CSS3 and vanilla JavaScript and integrated with the Groq AI API and Google Sheets, the platform was designed, developed, tested and deployed by the team within the SmartBridge SIP 2026 internship timeline, meeting all the functional requirements identified during Requirement Analysis.

10. Future Scope

- User accounts to track eligibility/credit history over time
- A secure backend layer to manage the Groq API key server-side
- Downloadable PDF reports of eligibility, credit and EMI results
- Multi-language support and a conversational chatbot interface
- Partnership integrations with NBFCs for pre-qualified loan referrals

11. Appendix

Source Code: [index.html](#), [eligibility.html](#), [credit-score.html](#), [emi.html](#), [tips.html](#), [css/style.css](#), [js/*.js](#), [google-apps-script.gs](#) (see GitHub repository).

Dataset Link: Not applicable - FinWise AI uses rule-based logic and live AI inference rather than a static training dataset; submitted user data is logged to Google Sheets.

GitHub Repository: <https://github.com/JaineshBharti5/finwise-ai>

Live Demo: <https://jaineshbharti5.github.io/finwise-ai>

Demo Video: <https://drive.google.com/file/d/1R7yStCQXAuEFCktQstJzgT8etC3Lp7Fr/view?usp=sharing>